

$$\int_0^{\infty} e^{-ax} \sin bx \, dx = \frac{b}{a^2 + b^2}$$

In the present case

$$f(-t) \cos u(-t) = f(t) \cos ut \quad \because f(-t) = f(t).$$

$$f(-t) \sin u(-t) = -f(t) \sin ut$$

$$\therefore \int_{-\infty}^{\infty} f(t) \cos ut \, dt = 2 \int_0^{\infty} f(t) \cos ut \, dt \quad \dots(3)$$

$$\int_{-\infty}^{\infty} f(t) \sin ut \, dt = 0. \quad \dots(4)$$

From (3)

$$\begin{aligned} \int_{-\infty}^{\infty} f(t) \cos ut \, dt &= 2 \int_0^{\infty} e^{-at} \cos ut \, dt \\ &= 2 \frac{a}{a^2 + u^2} \quad \dots(5) \end{aligned}$$

Writing (2) with the help of (4) and (5), we get

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \frac{2a \cos ux \, du}{a^2 + u^2} + 0$$

$$= \frac{2a}{\pi} \int_0^{\infty} \frac{\cos ux \, du}{a^2 + u^2}$$

$$\therefore f(x) = e^{-ax} = \frac{2a}{\pi} \int_0^{\infty} \frac{\cos ux \, du}{a^2 + u^2}$$

From which

$$\frac{\pi}{2a} e^{-ax} = \int_0^{\infty} \frac{\cos ux \, du}{a^2 + u^2}$$

Fourier integral of $f(x)$

$$= \frac{2a}{\pi} \int_0^{\infty} \frac{\cos ux \, du}{a^2 + u^2}$$

Ans.

Step (ii). Let $f(-x) = -f(x)$

Then

$$f(-t) \cos u(-t) = -f(t) \cos ut$$

$$f(-t) \sin u(-t) = f(t) \sin ut$$

As a result of which

$$\int_{-\infty}^{\infty} f(t) \cos ut \, dt = 0 \quad \dots(6)$$

$$\int_{-\infty}^{\infty} f(t) \sin ut \, dt = 2 \int_0^{\infty} f(t) \sin ut \, dt$$

$$= 2 \int_0^{\infty} e^{-at} \sin ut \, dt$$

$$= 2 \frac{u}{a^2 + u^2}$$

$$\therefore \int_{-\infty}^{\infty} f(t) \sin ut \, dt = \frac{2\pi}{a^2 + u^2} \quad \dots (7)$$

In view of (6) and (7); (2) becomes

$$f(x) = 0 + \frac{2}{\pi} \int_0^{\infty} \frac{u \sin ux \, du}{a^2 + u^2}$$

$$\text{i.e. } f(x) = e^{-ax} = \frac{2}{\pi} \int_0^{\infty} \frac{u \sin ux \, du}{u^2 + a^2}$$

From which

$$\frac{\pi}{2} e^{-ax} = \int_0^{\infty} \frac{u \sin ux}{u^2 + a^2} \, du$$

Fourier integral of $f(x)$

$$= \frac{2}{\pi} \int_0^{\infty} \frac{u \sin ux \, du}{u^2 + a^2}$$

Ans.

Exercise 2

Prove that

$$1. \, e^{-x} - e^{-2x} = \frac{6}{\pi} \int_0^{\infty} \frac{u \sin ux \, du}{(u^2 + 1)(u^2 + 4)}, \quad x \geq 0.$$

Hint. Use Fourier sine integral formula.

$$2. \, \int_0^{\infty} \frac{\cos\left(\frac{\pi}{2} u\right)}{1 - u^2} \, du = \frac{\pi}{2}.$$

Hint. Put $x = \frac{\pi}{2}$ in the result of Ex. 9.

3. Show, under conditions to be stated, that

$$\lim_{w \rightarrow \infty} \int_0^b f(x) \frac{\sin wx}{x} \, dx = \frac{1}{2} \pi f(+0). \quad (\text{B.H.U. 1965})$$

Hint. Refer solved Ex. 2.

The Fourier Transform

Part I—THE INFINITE FOURIER TRANSFORM

3.0. Def. The (infinite) Fourier sine transform of $F(x)$, $0 < x < \infty$, is denoted by $f_s(s)$ or $F_s\{F(x)\}$ and is defined as

$$f_s(s) = F_s\{F(x)\} = \int_0^{\infty} F(x) \sin sx \, dx.$$

The inverse Formula for infinite Fourier sine transform is given by

$$F(x) = F_s^{-1}\{f_s(s)\} = \frac{2}{\pi} \int_0^{\infty} f_s(s) \sin sx \, ds.$$

3.1. Def. The (infinite) Fourier cosine transform of $F(x)$, $0 < x < \infty$, is denoted by $f_c(s)$ or $F_c\{F(x)\}$ and is defined as,

$$f_c(s) = F_c\{F(x)\} = \int_0^{\infty} F(x) \cos sx \, dx.$$

The inverse formula for infinite Fourier cosine transform is given by

$$F(x) = F_c^{-1}\{f_c(s)\} = \frac{2}{\pi} \int_0^{\infty} f_c(s) \cos sx \, ds.$$

(Meerut 1969, 68)

Remark. Some authors take $\sqrt{(2/\pi)}$, $\sqrt{(2/\pi)}$ as the coefficients in the two equations [of Def. (3.0) and (3.1) both] instead of $2/\pi$ respectively

3.2. Def. The infinite Fourier transform of $F(x)$, $-\infty < x < \infty$, is denoted by $f(s)$ or $F\{F(x)\}$ and is defined as

$$f(s) = F\{F(x)\} = \int_{-\infty}^{\infty} F(x) e^{-isx} \, dx. \quad \dots(1)$$

The inverse formula for Fourier transform is given by

$$F(x) = F^{-1}\{f(s)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(s) e^{isx} ds. \quad \dots(2)$$

Remark. (1) Some authors take the coefficients in (1) and (2) as $\sqrt{(2/\pi)}$ each, instead of 1 and $1/2\pi$.

(2) Some authors define Fourier transform as follows :

$$f(s) = F\{F(x)\} = \int_{-\infty}^{\infty} F(x) e^{isx} dx$$

and $F(x) = F^{-1}\{f(s)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(s) e^{-isx} ds.$

But this definition and the definition given above lead to the same conclusion.

Solved Problems

Problem 1. Find the Fourier transform of

$$f(x) = \begin{cases} x, & |x| \leq a \\ 0, & |x| > a \end{cases} \quad (\text{Kanpur 1985, 80, 79})$$

Solution. Given that

$$f(t) = \begin{cases} t, & -a \leq t \leq a \\ 0, & |t| > a \end{cases}$$

$$F\{f(t)\} = \int_{-\infty}^{\infty} e^{-ist} f(t) dt$$

$$= \int_{-\infty}^{-a} e^{-ist} f(t) dt + \int_{-a}^a e^{-ist} f(t) dt + \int_a^{\infty} e^{-ist} f(t) dt$$

$$= \int_{-\infty}^{-a} e^{isy} f(y) (-dy) + \int_{-a}^a e^{-ist} t dt + \int_a^{\infty} e^{-ist} 0 dt.$$

$$= \int_a^{\infty} e^{isy} f(-y) dy + \left[\frac{e^{-ist} t}{-is} + \frac{e^{-ist}}{s^2} \right]_{t=-a}^a + 0$$

$$= \int_a^{\infty} e^{isy} 0 dy - \frac{a}{is} (e^{-isa} + e^{isa}) + \frac{1}{s^2} (e^{-isa} - e^{isa})$$

$$= 0 + \frac{ia}{s} \cdot 2 \cos sa + \frac{1}{s^2} \cdot (-2i \sin sa)$$

$$= \frac{2i}{s^2} (as \cos sa - \sin sa)$$

Ans.

Example 2. Find the Fourier transform of

$$f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$$

(Kanpur 1981, 79, 70; Meerut 1991
Agra '81; Roorkee M.E. 1967)

Solution. Given that

$$f(x) = \begin{cases} 1, & \text{for } |x| < a, \text{ i.e., for } -a < x < a \\ 0, & \text{for } |x| > a. \end{cases}$$

$F\{f(x)\}$

$$\begin{aligned} &= \int_{-\infty}^{\infty} e^{-isx} f(x) dx \\ &= \int_{-\infty}^{-a} e^{-isx} f(x) dx + \int_{-a}^a e^{-isx} f(x) dx + \int_a^{\infty} e^{-isx} f(x) dx \\ &= \int_{\infty}^a -e^{isy} f(-y) dy + \int_{-a}^a e^{-isx} \cdot 1 \cdot dx + \int_0^{\infty} e^{-isx} \cdot 0 \cdot dx \\ &= \int_a^{\infty} e^{isy} f(-y) dy + \left(\frac{e^{-isx}}{-is} \right)_{x=-a}^a + 0 \\ &= \int_a^{\infty} e^{isy} \cdot 0 \cdot dy + \frac{e^{isa} - e^{-isa}}{is} + 0 = 0 + \frac{2}{s} \sin sa = \frac{2}{s} \sin sa. \end{aligned}$$

Ans.

Problem 3. Find sine and cosine transform of $x^n e^{-ax}$.

(Kanpur 1981)

Solution. We know that

$$\int_0^{\infty} e^{-ax} \cos px \, dx = \frac{a}{a^2 + p^2} \quad \dots(1)$$

$$\int_0^{\infty} e^{-ax} \sin px \, dx = \frac{p}{p^2 + a^2} \quad \dots(2)$$

Let $\bar{f}_c(p)$ and $\bar{f}_s(p)$ be the required cosine and sine transform respectively. Differentiating (1) n times w.r.t. a , we get

$$\int_0^{\infty} e^{-ax} (-x)^n \cos px \, dx = \frac{d^n}{da^n} \left\{ \frac{a}{a^2 + p^2} \right\}$$

$$\text{or } (-1)^n \int_0^{\infty} e^{-ax} x^n \cos px \, dx = (-1)^n \frac{n!}{p^{n+1}} \{ \cos(n+1)\theta \} \quad (\sin \theta)^{n+1}$$

$$\text{where } \tan \theta = p/a$$

$$\text{or } \bar{f}_c(p) = n! \{ \cos(n+1)\theta \} / (a^2 + p^2)^{(n+1)/2} \quad \dots(2)$$

Differentiating (2) w.r.t. a , n times we get

$$\int_0^{\infty} e^{-ax} \{ \sin px \} (-x)^n \, dx = p \frac{d^n}{da^n} \left(\frac{1}{a^2 + p^2} \right)$$

$$\text{or } (-1)^n \int_0^{\infty} e^{-ax} x^n \sin px \, dx = p (-1)^n \frac{n!}{p^{n+2}} \{ \sin(n+1)\theta \} \quad (\sin \theta)^{n+1}$$

or $f_s(p) = n! \{ \sin(n+1)\theta \} / (a^2 + p^2)^{(n+1)/2} \dots (4)$

From (3) and (4), the required result follows.

Problem 3. Find the Fourier transform of $f(x)$ defined by

$$f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$$

and hence evaluate

$$(i) \int_{-\infty}^{\infty} \frac{\sin sa \cos sx}{s} ds, \quad (ii) \int_0^{\infty} \frac{\sin s}{s} ds.$$

(Agra 1973, Meerut 73; Kanpur 70; Roorkee 66)

Solution. First Part. To determine $F\{f(x)\}$.

For this, see the above solved Ex. 2.

Second part. Let $F\{f(x)\} = \bar{f}(s)$.

We know that if

$$\bar{f}(s) = F\{f(x)\} = \int_{-\infty}^{\infty} f(x) e^{-isx} dx$$

then $f(x) = F^{-1}\{\bar{f}(s)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \bar{f}(s) e^{isx} ds.$

$$\therefore \int_{-\infty}^{\infty} \bar{f}(s) e^{isx} ds = 2\pi f(x) = \begin{cases} 2\pi & \text{if } |x| < a \\ 0 & \text{if } |x| > a. \end{cases}$$

But $\bar{f}(s) = \frac{2 \sin sa}{s}$, by first part.

$$\therefore \int_{-\infty}^{\infty} \frac{2 \sin sa}{s} (\cos sx + i \sin sx) ds = \begin{cases} 2\pi & \text{if } |x| < a \\ 0 & \text{if } |x| > a \end{cases}$$

or $\int_{-\infty}^{\infty} \frac{\sin sa \cos sx}{s} ds + i \int_{-\infty}^{\infty} \frac{\sin sa \sin sx}{s} ds = \begin{cases} \pi, & |x| < a \\ 0, & |x| > a \end{cases}$

Equating real parts of both sides,

or $\int_{-\infty}^{\infty} \frac{\sin sa \cos sx}{s} ds = \begin{cases} \pi & \text{if } |x| < a \\ 0 & \text{if } |x| > a \end{cases}$

Ans.

Third Part. If $x=0, a=1$, then the second part gives.

$$\int_{-\infty}^{\infty} \frac{\sin s}{s} ds = \pi. \quad \text{For } |x| < a \Rightarrow 0 < 1 \text{ and } \cos 0 = 1].$$

or $2 \int_0^{\infty} \frac{\sin s}{s} ds = \pi$

or $\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}.$

Ans.

Problem 4. Find the Fourier sine transform of

$$f(x) = \begin{cases} x & \text{for } 0 < x < 1 \\ 2-x & \text{for } 1 < x < 2 \\ 0 & \text{for } x > 2 \end{cases}$$

(Kanpur 1988, 79, 77)

Solution. $F_s \{f(x)\} = \int_0^\infty f(x) \sin sx \, dx$

$$= \int_0^1 f(x) \sin sx \, dx + \int_1^2 f(x) \sin sx \, dx + \int_2^\infty f(x) \sin sx \, dx$$

$$= \int_0^1 x \sin sx \, dx + \int_1^2 (2-x) \sin sx \, dx + \int_2^\infty 0 \sin sx \, dx$$

$$= \left[-\frac{x \cos sx}{s} + \frac{\sin sx}{s^2} \right]_{x=0}^1 + \left[-\frac{2 \cos sx}{s} + \frac{\sin sx}{s^2} \right]_{x=1}^2 - \left[-\frac{x \cos sx}{s} + \frac{\sin sx}{s^2} \right]_{x=1}^2 + 0$$

$$= \left(-\frac{\cos s}{s} + \frac{\sin s}{s^2} \right) - \frac{2}{s} \cos 2s + \frac{2}{s} \cos s$$

$$+ \frac{2 \cos 2s - \cos s}{s} - \left(\frac{\sin 2s - \sin s}{s^2} \right)$$

$$= \frac{2 \sin s - 2 \sin s \cos s}{s^2} = \frac{2(1 - \cos s) \sin s}{s^2}.$$

Ans.

Change of scale property.

Theorem 1. If $f(s)$ is the Fourier transform of $F(x)$, then $\frac{1}{a} f\left(\frac{s}{a}\right)$ is the Fourier transform of $F(ax)$.

Proof. To prove that $F\{F(ax)\} = \frac{1}{a} f\left(\frac{s}{a}\right)$.

$$\therefore F\{F(x)\} = \int_{-\infty}^\infty e^{-isx} F(x) \, dx = f(s)$$

$$\therefore F\{F(ax)\} = \int_{-\infty}^\infty e^{-isx} F(ax) \, dx$$

$$= \int_{-\infty}^\infty e^{-\frac{ist}{a}} F(t) \frac{dt}{a}, \quad ax = t$$

$$= \frac{1}{a} \int_{-\infty}^\infty e^{-i\left(\frac{s}{a}\right)t} F(t) \, dt$$

$$= \frac{1}{a} f\left(\frac{s}{a}\right) \text{ by definition,} \quad \textbf{Proved.}$$

Theorem 2. If $f_c(s)$ is the Fourier cosine transform of $F(x)$, then show that Fourier cosine transform of $F\left(\frac{x}{a}\right)$ is $a f_c(as)$.

Proof. $F_c\left\{F\left(\frac{x}{a}\right)\right\} = \int_0^\infty F\left(\frac{x}{a}\right) \cos sx \, dx$

$$\begin{aligned}
 &= \int_0^{\infty} F(t) \cos(sat) \, adt, \quad \frac{x}{a} = t \\
 &= \int_0^{\infty} aF(t) \cos(sat) \, dt = a f_c(as)
 \end{aligned}$$

Theorem 3. To prove that

$$F_s \left\{ F \left(\frac{x}{a} \right) \right\} = a f_s(as).$$

The proof is left as an exercise for students.

Theorem 4. Shifting property. If $f(s)$ is the Fourier transform of $F(x)$, then

$e^{-ias} f(s)$ is the Fourier transform of $F(x-a)$.

$$\begin{aligned}
 \text{Proof. } F\{F(x-a)\} &= \int_{-\infty}^{\infty} e^{-isx} F(x-a) \, dx && x-a=t \\
 &= \int_{-\infty}^{\infty} e^{-is(a+t)} F(t) \, dt, && x-a=t \\
 &= \int_{-\infty}^{\infty} e^{-isa} e^{-ist} F(t) \, dt = e^{-isa} \int_{-\infty}^{\infty} e^{-ist} F(t) \, dt \\
 &= e^{-isa} f(s) && \text{Proved.}
 \end{aligned}$$

Theorem 5. Modulation Theorem. If $F(x)$ has the Fourier transform $f(s)$, then $F(x) \cos ax$ has the Fourier transform

$$\frac{1}{2} f(s-a) + \frac{1}{2} f(s+a). \quad (\text{Meerut B Sc. Hons. 1972})$$

Proof. Recall that $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$.

$$\begin{aligned}
 F\{F(x) \cos ax\} &= \int_{-\infty}^{\infty} e^{-isx} F(x) \cos ax \, dx \\
 &= \int_{-\infty}^{\infty} e^{-isx} F(x) \frac{e^{iax} + e^{-iax}}{2} \, dx \\
 &= \frac{1}{2} \int_{-\infty}^{\infty} \left[e^{-i(s-a)x} F(x) + e^{-i(s+a)x} F(x) \right] \, dx \\
 &= \frac{1}{2} \left[\int_{-\infty}^{\infty} e^{-i(s-a)x} F(x) \, dx + \int_{-\infty}^{\infty} e^{-i(s+a)x} F(x) \, dx \right] \\
 &= \frac{1}{2} [f(s-a) + f(s+a)] && \text{Ans.}
 \end{aligned}$$

Remember the results of the following definite integrals :

$$(1) \int_0^{\infty} e^{-x^2} \, dx = \frac{\sqrt{\pi}}{2}$$

$$(2) \Gamma(n) = a^n \int_0^{\infty} e^{-ax} x^{n-1} \, dx$$

$$(3) \int_0^{\infty} e^{-ax} \cos bx \, dx = \frac{a}{a^2 + b^2}$$

$$(4) \int_0^{\infty} \frac{\sin ax}{x} dx = \frac{\pi}{2} \text{ if } a > 0.$$

Problems related to integral equations.

Problem 5. Solve the integral equation

$$\int_0^{\infty} f(x) \cos \lambda x dx = e^{-\lambda}$$

(Kanpur 1990, Kashmir 64)

$$\text{Solution. } \int_0^{\infty} f(x) \cos sx dx = e^{-s} \quad \dots(1)$$

$$\text{By def., } F_c \{f(x)\} = \int_0^{\infty} f(x) \cos sx dx = \bar{f}_c(s) \quad \dots(2)$$

$$\text{and } F_c^{-1} \{\bar{f}_c(s)\} = f(x) = \frac{2}{\pi} \int_0^{\infty} \bar{f}_c(s) \cos sx ds \quad \dots(3)$$

Equating (1) to (2), $\bar{f}_c(s) = e^{-s}$

Using this in (3), we obtain

$$\begin{aligned} f(x) &= \frac{2}{\pi} \int_0^{\infty} e^{-s} \cos sx ds \\ &= \frac{2}{\pi} \int_0^{\infty} e^{-sx} \cos xs ds = \frac{2}{\pi} \cdot \frac{1}{1+x^2} \\ &\quad \left[\text{For } \int_0^{\infty} e^{-ax} \cos bx dx = \frac{a}{a^2+b^2} \right] \\ \therefore f(x) &= \frac{2}{\pi(1+x^2)} \end{aligned}$$

Problem 6. Show that

$$\int_0^{\infty} \frac{\cos \lambda x}{\lambda^2+1} d\lambda = \frac{\pi}{2} e^{-x}, x \geq 0. \quad (\text{Roorkee M. E. 67})$$

Solution. We know that

$$\int_0^{\infty} e^{-ax} \cos bx = \frac{a}{a^2+b^2}$$

$$\text{From which } \int_0^{\infty} e^{-x} \cos \lambda x dx = \frac{1}{1+\lambda^2}.$$

Comparing this with the equation,

$$F_c \{f(x)\} = \bar{f}_c(\lambda) = \int_0^{\infty} f(x) \cos \lambda x dx \quad \dots(1)$$

$$\text{We get } \bar{f}_c(\lambda) = \frac{1}{1+\lambda^2}, f(x) = e^{-x}$$

The inverse formula corresponding to (1) is

$$F_c^{-1} \{\bar{f}_c(\lambda)\} = f(x) = \frac{2}{\pi} \int_0^{\infty} \bar{f}_c(\lambda) \cos \lambda x d\lambda$$